

SAMPLING AND DECENTRALIZED FIXED MODES

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Abstract

The effect of sampling on decentralized fixed modes is analyzed. Decentralized fixed modes are classified as being "structured" and "unstructured". It is shown that for almost all sampling intervals the unstructured decentralized fixed modes will no longer exist in a system after the introduction of sampling. This implies that a continuous-time system with unstable decentralized fixed modes may still be stabilized with digital controllers for almost all sampling intervals.

1. Introduction

An important question in the control of decentralized systems is the existence or non-existence of decentralized fixed modes. It has been shown [1] that a decentralized linear system is stabilizable (using time-invariant possibly dynamic compensators) if and only if the decentralized fixed modes are stable.

Recent results reported by Anderson and Moore [2] and Wang [3] show that time-varying controllers may be introduced to drive the states to zero, in spite of the existence of decentralized fixed modes.

In the present paper sampled systems with zero-order-holds at each input are analyzed. The question regarding stabilization in spite of unstable decentralized fixed modes in the continuous-time model is clarified. It is shown that, for almost all sampling intervals, the continuous-time decentralized fixed modes will not appear in the discrete-time model. An explanation of this phenomenon may be based on the different effect of sampling on poles and zeros (see for example [4]). Thus, recent results on transfer-function based analyses of decentralized fixed modes of Özgüner et al. [5] and Davison and Wang [6] are utilized in the sequel.

2. Preliminaries

Consider a plant consisting of v agents described by:

$$\dot{x} = Ax + \sum_{i=1}^v B_i u_i \quad (1)$$

$$y_i = C_i x, \quad i=1,2,\dots,v$$

where $x \in \mathbb{R}^n$, $u_i \in \mathbb{R}^{m_i}$, $y_i \in \mathbb{R}^{r_i}$, $i=1,2,\dots,v$. Let

$$B = (B_1, B_2, \dots, B_v)$$

$$C = \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_v \end{pmatrix}$$

and assume that (C,A,B) is controllable and observable, and that A has n distinct eigenvalues λ_i , $i=1,2,\dots,n$.

Let a coordinate transformation be applied to (1) to obtain:

$$\dot{\bar{x}} = \lambda \bar{x} + \sum_{i=1}^v \bar{B}_i u_i \quad (2)$$

$$y_i = \bar{C}_i \bar{x}, \quad i=1,2,\dots,v$$

where λ is a diagonal matrix given by

$$\lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

The following definition is made:

Definition

Let λ_i be a decentralized fixed mode [1] of (1). Then λ_i is said to be a structured decentralized fixed mode of (1) if λ_i is a decentralized fixed mode of (2) for all non zero elements

$\bar{B}_i, \bar{C}_i, i=1,2,\dots,v$; otherwise λ_j is said to be a unstructured decentralized fixed mode of (1). (see also [5])

Assume now that (1) contains $P_u > 0$ unstructured decentralized fixed modes given by $\lambda_j, j=1,2,\dots,P_u$, and has $P_s > 0$ structured decentralized fixed modes given by $\lambda_j, j=1,2,\dots,P_s$ where $\lambda_i \neq \lambda_j, i=1,2,\dots,P_u, j=1,2,\dots,P_s$.

The following results are obtained:

Lemma 1

Consider the plant (1) with $v > 3$ control agents, and assume that (1) has P_u unstructured decentralized fixed modes and P_s structured decentralized fixed modes. Then this is true if and only if the following v plants with $v-1$ control agents have the property that (i) the union of their unstructured decentralized fixed modes is equal to the P_u unstructured decentralized fixed modes of (1), and (ii) the union of their structured decentralized fixed modes is equal to the P_s structured decentralized fixed modes of (1):

$$(i) \left\{ \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ \vdots \\ C_v \end{bmatrix}, A, [(B_1, B_2), B_3, \dots, B_v] \right\}$$

$$(ii) \left\{ \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \\ \vdots \\ C_v \end{bmatrix}, A, [(B_2, B_3), B_4, \dots, B_v] \right\}$$

$$(v-1) \left\{ \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_{v-1} \\ C_v \end{bmatrix}, A, [B_1, B_2, \dots, (B_{v-1}, B_v)] \right\}$$

$$(vi) \left\{ \begin{bmatrix} C_1 \\ \vdots \\ C_{v-3} \\ (C_{v-2}) \\ C_v \\ C_{v-1} \end{bmatrix}, A, [B_1, \dots, B_{v-3}, (B_{v-2}, B_v), B_{v-1}] \right\}$$

Proof

See Appendix I.

Remark

This lemma states that problems of characterizing the decentralized fixed modes of a v agent control system reduces to the problem of characterizing decentralized fixed modes of a family of 2 control agent systems.

Lemma 2

Consider the plant (1) which has P_u unstructured decentralized fixed modes and P_s structured decentralized fixed modes; then the system:

$$\dot{x} = Ax + \sum_{i=1}^v B_i \rho_i v_i \quad (3)$$

$$z_i = g_i C_i x, \quad i=1,2,\dots,v$$

where $\rho_i \in \mathbb{R}^{m_i \times 1}$, $g_i \in \mathbb{R}^{1 \times r_i}$, $i=1,2,\dots,v$ has the P_u unstructured decentralized fixed modes and the same structured decentralized fixed modes as (1) for almost all [7] $\rho_i \in \mathbb{R}^{m_i \times 1}$, $g_i \in \mathbb{R}^{1 \times r_i}$, $i=1,2,\dots,v$.

Proof.

See Appendix II.

Remark

This result states that for almost all linear combinations of a control agents' inputs, and outputs, there is no loss of generality in assuming that each control agent of (1) possesses only a single-input v_i a single-output z_i , $i=1,2,\dots,v$ with respect to the system's decentralized fixed modes.

Discussion

Lemma 1 and 2 together imply that the problem of characterizing the decentralized fixed modes of a v control agent system with m_i, r_i

inputs, outputs respectively, $i=1,2,\dots,v$ reduces to the problem of characterizing the decentralized fixed modes of a 2 control agent system with $m_i=r_i$, $i=1,2$.

The following result is now obtained for characterizing the structured and unstructured decentralized fixed modes of (1) for the case when $v=2$ and $m_i=r_i=1$, $i=1,2$.

$$\dot{x} = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} x + \begin{bmatrix} b_1^1 \\ b_1^2 \\ b_2^1 \\ b_2^2 \\ \vdots \\ b_n^1 \\ b_n^2 \end{bmatrix} u_1 + \begin{bmatrix} b_1^2 \\ b_1^1 \\ b_2^2 \\ b_2^1 \\ \vdots \\ b_n^2 \\ b_n^1 \end{bmatrix} u_2 \quad (4)$$

$$y_1 = (c_1^1, c_2^1, \dots, c_n^1)x$$

$$y_2 = (c_1^2, c_2^2, \dots, c_n^2)x$$

Lemma 3

Given the system (1) with $v=2$, $m_i=r_i=1$, $i=1,2$, then λ_1 is a structured decentralized fixed of (1) if and only if either condition (a) or condition (b) given below holds:

$$(a) \quad b_1^1=0 \text{ and } c_1^2=0 \text{ and } b_i^1 c_i^2=0, \quad i=2,3,\dots,n$$

$$(b) \quad b_1^2=0 \text{ and } c_1^1=0 \text{ and } b_i^2 c_i^1=0, \quad i=2,3,\dots,n$$

and λ_1 is a unstructured decentralized fixed mode of (1) if and only if conditions (a), (b) do not hold and either condition (c) or condition (d) given below holds:

$$(c) \quad b_1^1=0 \text{ and } c_1^2=0 \text{ and}$$

$$(c_2^2, c_3^2, \dots, c_n^2) \text{diag}^{-1}(\lambda_2 - \lambda_1, \lambda_3 - \lambda_1, \dots, \lambda_n - \lambda_1) \begin{bmatrix} b_2^1 \\ b_3^1 \\ \vdots \\ b_n^1 \end{bmatrix} = 0$$

$$(d) \quad b_1^2=0 \text{ and } c_1^1=0 \text{ and}$$

$$(c_2^1, c_3^1, \dots, c_n^1) \text{diag}^{-1}(\lambda_2 - \lambda_1, \lambda_3 - \lambda_1, \dots, \lambda_n - \lambda_1) \begin{bmatrix} b_2^2 \\ b_3^2 \\ \vdots \\ b_n^2 \end{bmatrix} = 0$$

Proof

See Appendix III.

The following result follows after simple algebraic manipulation:

Corollary 1

Conditions (c), (d) of Lemma 3 are equivalent to the following conditions (c)', (d)' respectively:

$$(c)' \quad b_1^1=0 \text{ and } c_1^2=0 \text{ and}$$

$$\sum_{i=2}^n c_i^2 b_i^1 \prod_{\substack{j=2 \\ j \neq i}}^n (\lambda_j - \lambda_1) = 0$$

$$(d)' \quad b_1^2=0 \text{ and } c_1^1=0 \text{ and}$$

$$\sum_{i=2}^n c_i^1 b_i^2 \prod_{\substack{j=2 \\ j \neq i}}^n (\lambda_j - \lambda_1) = 0$$

The following result therefore follows from Lemma 3 and Corollary 1:

Lemma 4

Given the system (1) with $v=2$, $m_i=r_i=1$, $i=1,2$, then λ_1 is a unstructured decentralized fixed mode of (1) if and only if either condition (a)' or condition (b)' given below holds:

$$(a)' \quad b_1^1=0 \text{ and } c_1^2=0 \text{ and}$$

$$\sum_{i=2}^n c_i^2 b_i^1 \prod_{\substack{j=2 \\ j \neq i}}^n (\lambda_j - \lambda_1) = 0, \text{ where } c_i^2 b_i^1 \neq 0$$

for at least one $i \in [2, 3, \dots, n]$.

(b) $b_1^2 = 0$ and $c_1^1 = 0$ and

$$\sum_{i=2}^n c_i^1 b_i^2 \prod_{j=2}^n (\lambda_j - \lambda_1) = 0, \text{ where } c_i^1 b_i^2 \neq 0 \text{ for } i \neq 1$$

least one $i \in [2, 3, \dots, n]$.

3. Main Results

Assume now that a digital controller is used to control (1) with a constant sampling interval of $h > 0$ and a zero-order hold is used. Then the resultant sampled system may be described as follows:

$$\dot{x}_{t+h} = e^{\lambda h} x_t + r \sum_{i=1}^v \bar{b}_i u_i \quad (5)$$

$$y_{i_t} = \bar{c}_i x_{i_t}, \quad i=1, 2, \dots, v$$

where

$$r = \Delta \text{diag} (r_1, r_2, \dots, r_n)$$

$$\text{where if } \lambda_i \neq 0, r_i = \frac{\Delta}{\lambda_i} \frac{e^{\lambda_i h} - 1}{h}, \text{ and if } \lambda_i = 0,$$

$$r_i = \Delta h, \text{ and where } \lambda, \bar{b}_i, \bar{c}_i \text{ are defined in (2).}$$

Assume now that (1) contains $P_u > 0$ unstructured decentralized fixed modes and $P_s > 0$ structured decentralized fixed modes.

The following definition is made:

Definition

$$\text{Given } f(h) \triangleq \sum_{i=2}^n \theta_i f_i(h)$$

where

$$f_i(h) \triangleq g_i(h) \prod_{j=2}^n (e^{\lambda_j h} - \bar{g}_1(h)), \quad i=2, 3, \dots, n$$

and

$$g_i(h) \triangleq \begin{cases} e^{\lambda_i h} - 1 & \text{if } \lambda_i \neq 0 \\ h & \text{if } \lambda_i = 0 \end{cases}$$

and

$$\bar{g}_1(h) \triangleq \begin{cases} e^{\lambda_1 h} & \text{if } \lambda_1 \neq 0 \\ h & \text{if } \lambda_1 = 0 \end{cases}$$

where $\lambda_i \in \mathbb{C}$, $i=1, 2, \dots, n$ are symmetric and distinct, then if either of the following conditions hold:

$$(i) \quad f(h) \neq 0, \quad \forall h > 0$$

$$(ii) \quad \text{Given any } h^* > 0, \text{ there exists only a finite number of } h \in (0, h^*] \text{ such that } f(h) = 0$$

$f(h) \neq 0$ is said to be true for almost all $h > 0$.

The following result is now obtained:

Theorem 1

Given the system (1), assume that (1) contains $P_u > 0$ unstructured decentralized fixed modes and $P_s > 0$ structured decentralized fixed modes λ_j , $j=1, 2, \dots, P_s$; then the sampled system (5) contains P_s structured decentralized fixed modes $e^{\lambda_j h}$, $j=1, 2, \dots, P_s$, $\forall h > 0$, and no unstructured decentralized fixed modes for almost all $h > 0$.

Proof

See Appendix IV.

Discussion

This theorem states that if a plant (1) contains no structured decentralized fixed modes, then a digital controller can be used to stabilize, pole-assign etc. the plant for almost all sampling intervals. This is the case even though the plant (1) may contain unstructured decentralized fixed modes (which implies that there exists no continuous linear time-invariant controller to "shift" these decentralized modes [1]).

The result obtained is consistent with the results reported by both Anderson and Moore [2] and by Wang [3]. The issue becomes clearer when decentralized fixed modes are viewed as pole-zero cancellations of a specific kind in the decentralized system. The distribution of poles and zeros that cancel to create decentralized fixed modes are clearly shown in [5]. Thus, the different effect sampling has on poles and zeros [4] result in the cancellations not to occur in the model of the sampled system.

4. Numerical Examples

Example No. 1

Consider the system:

$$\begin{aligned} \dot{x} &= \begin{pmatrix} -1 & 0 & -1 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{pmatrix} x + \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} u \\ y &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} x \end{aligned} \quad (6)$$

This system is controllable and observable with a non-structured decentralized fixed mode at -2. When this system is sampled with a sampling time interval of $h>0$, the resultant system is described by:

$$x_{t+h} = \begin{pmatrix} e^{-h} & 0 & \frac{1}{2}(e^{-h} - e^{-3h}) \\ 0 & e^{-2h} & 0 \\ 0 & 0 & e^{-3h} \end{pmatrix} x_t + \begin{pmatrix} \frac{2}{3} & -\frac{1}{2}e^{-h} & -\frac{1}{6}e^{-3h} \\ 0 & \frac{1}{2} & -\frac{1}{2}e^{-2h} \\ \frac{1}{3} & -\frac{1}{3} & e^{-3h} \end{pmatrix} u_t \quad (7)$$

$$y_t = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} x_t$$

and this system has a decentralized fixed mode present if and only if

$$f(h) \triangleq e^{-2h} + e^{-4h} - \frac{1}{4}(e^{-h} + e^{-3h} + e^{-5h}) = 0 \quad (8)$$

Since $f(h) < 0$, $\forall h > 0$ this implies that the sampled system (7) has no decentralized fixed modes present $\forall h > 0$ which is consistent with Theorem 1.

Example No. 2

Consider the system:

$$\dot{x} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{pmatrix} x + \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} u$$

$$y = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} x$$

This system is controllable and observable with a structured decentralized fixed mode at -2. It may be verified that when the system is sampled, the resultant system has a decentralized fixed mode at e^{-2h} , which is consistent with Theorem 1.

Example No. 3

Consider the unstable controllable, observable system:

$$\dot{x} = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix} x + \begin{pmatrix} 0 & 1 \\ -3 & 0 \end{pmatrix} u$$

$$y = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} x + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} u \quad (10)$$

which has a non-structured decentralized fixed mode at +1. When this system is sampled, the resultant system is described by:

$$x_{t+h} = \begin{pmatrix} e^h & 0 \\ 0 & e^{-2h} \end{pmatrix} x_t + \begin{pmatrix} 0 & e^h - 1 \\ -\frac{3}{2}(1 - e^{-2h}) & 0 \end{pmatrix} u_t$$

$$y_t = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} x_t + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} u_t \quad (11)$$

and this system has a decentralized fixed mode e^h present if and only if h satisfies the following equation:

$$-\frac{3}{2} + \frac{1}{2}e^{-2h} + 2 = 0 \quad (12)$$

Since (12) has no solution for $h > 0$, this implies that the sampled system (11) has no decentralized fixed modes present $\forall h > 0$.

It may be verified in fact that the discrete controller

$$u_{t_1} = y_{t_1}, \quad u_{t_2} = -y_{t_2}, \quad h = 1/2 \quad (13)$$

stabilizes the continuous system (10) in spite of the fact that there exists no linear time-invariant continuous controller which stabilizes the system (10).

Conclusions

In this paper the effect of sampling on decentralized fixed modes was analyzed. It was shown that for almost all sampling intervals the decentralized system with zero-order holds will have no unstructured decentralized fixed modes, even though the continuous-time system may originally have some unstructured decentralized fixed modes. This result implies that digital controllers may be used to stabilize the decentralized system. The result also clarifies recently published examples and shows that they fall into the class of controllers considered above.

References

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Appendices, I, II, and III are omitted because of space restrictions.

Appendix IV

The following result immediately follows on using the same development as used to establish Lemma 4:

Lemma 5

Consider the system (4) with $v=2$, $m_i=r_i=1$, $i=1,2$ in which λ_1 is an unstructured decentralized fixed mode; then $e^{\lambda_1 h}$ is a decentralized fixed mode of (5) if and only if either condition (i) or condition (ii), given below holds:

if $b_1=0$ and $c_1=0$ and $\sum_{i=2}^n c_i b_i \frac{1}{\lambda_i} (\lambda_j - \lambda_1) = 0$.

where $c_i b_i \neq 0$ for at least one $i \in [2, 3, \dots, n]$, then condition (i) is:

for $\lambda_1 \neq 0$:

$$(i) \sum_{i=2}^n c_i b_i \frac{1}{\lambda_i} g_i(h) \frac{1}{\pi} (e^{\lambda_j h} - e^{\lambda_1 h}) = 0$$

where

$$g_i(h) = \begin{cases} \frac{1}{\lambda_i} (e^{\lambda_j h} - e^{\lambda_1 h}) & \text{if } \lambda_j \neq 0 \\ h & \text{if } \lambda_j = 0 \end{cases}$$

and for $\lambda_1 = 0$:

$$(i) \sum_{i=2}^n c_i b_i \frac{1}{\lambda_i} \frac{e^{\lambda_j h} - 1}{\lambda_j} \frac{1}{\pi} (\lambda_j - \lambda_i) = 0, \text{ where}$$

$c_i b_i \neq 0$ for at least one $i \in [2, 3, \dots, n]$, then

condition (ii) is:

$$(ii) \text{ The same as condition (i) with } \frac{1}{\lambda_i} \frac{e^{\lambda_j h} - 1}{\lambda_j} \text{ replacing the term } \frac{1}{\lambda_i}.$$

It is observed now that conditions (i) or (ii) in Lemma 5 may be written as:

$$f(h) = \sum_{i=2}^n \theta_i f_i(h) = 0 \quad (1d)$$

where $\theta_i \neq 0$ for at least one $i \in [2, 3, \dots, n]$, and where

$$f_i(h) = \begin{cases} g_i(h) \frac{1}{\pi} (e^{\lambda_j h} - e^{\lambda_1 h}) & \text{if } \lambda_j \neq 0 \\ \frac{1}{\lambda_i} \sum_{j=2}^n (e^{\lambda_j h} - e^{\lambda_1 h}) & \text{if } \lambda_j = 0 \end{cases}, i=2, 3, \dots, v \quad (2d)$$

Now since $\lambda_1, i=1, 2, \dots, n$ are assumed to be all distinct, this implies that $f_i(h), i=2, 3, \dots, n$, are independent functions of h , which implies that

$$\sum_{i=2}^n \theta_i f_i(h) = 0, \quad \forall h \quad (3d)$$

if and only if $\theta_i = 0, i=2, 3, \dots, n$. But since $\theta_i \neq 0$ for at least one $i \in [2, 3, \dots, n]$, this implies that $f(h) \neq 0$ for almost any $h > 0$. The following result therefore follows:

Lemma 6

Consider the system (4) with $v=2$, $m_i=r_i=1$, $i=1,2$ in which λ_1 is an unstructured decentralized fixed mode; then $e^{\lambda_1 h}$ is not a decentralized fixed mode of (5) for almost all h .

The proof of Theorem 1 now follows immediately from Lemma 6, on noting that Lemma 1 and Lemma 2 also directly apply to the sampled system (5).